

Prove statements 15 and 16 directly
from the definition of divisibility.

4.4.16. For all integers a , b , and c , if $a|b$ and $a|c$ then

$$a|(b + c)$$

Suppose a , b , and c are integers.

We must show that if b is divisible by a , and c
is divisible by a , then the sum of b and c is
divisible by a .

$$\forall a, b, c \in \mathbb{Z}, a|b \wedge a|c \rightarrow a|(b + c)$$

The definition of divisibility according to Epp:

" n is divisible by $d \leftrightarrow n = d * k$, for some integer

k , and $d \neq 0$ "

Let $a, b, c \in \mathbb{Z}$

$$a|b \leftrightarrow b = ak, \text{ for some } k \in \mathbb{Z}$$

$$a|c \leftrightarrow c = aj, \text{ for some } j \in \mathbb{Z}$$

$$a \neq 0$$

$$(b + c) = (ak + aj) = a(k + j)$$

$$a(k + j) / a = k + j$$

$$(k + j) \in \mathbb{Z}$$

Solution: Therefore, if $a|b$ and $a|c$, then $a|(b + c)$